Mobile Application Development Laboratory

Real-Time Screen Fatigue Monitoring System - Dashboard App

Register No. : 230701011
Name : Aboorvan S B
Guide Name: Dr Deepa B
Designation and Department: Student (III Year) & Computer Science Engineering

RAJALAKSHMI ENGINEERING COLLEGE

Introduction

Real-Time Screen Fatigue Monitoring System

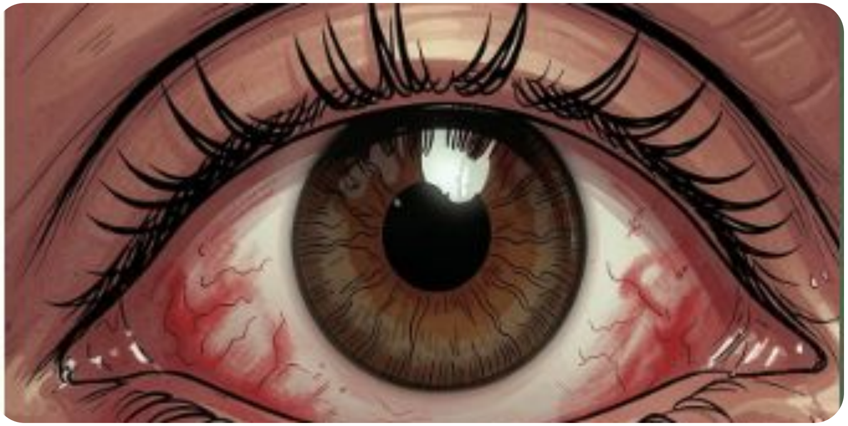
In today's digital environment, users spend long hours on laptops and mobile devices, which can lead to eye strain and fatigue. Continuous screen exposure reduces productivity and affects health.

The Project Solution

This project introduces a mobile-based fatigue monitoring system that uses computer vision and a Flutter application to track eye behavior and provide real-time alerts.

Motivation

- Detect fatigue early
- Improve user awareness
- Live monitoring via mobile dashboard



Problem Statement

User Challenges

- Excessive screen time causes eye strain
- Users often ignore fatigue symptoms
- No real-time monitoring available
- Lack of instant alerts and visual feedback

Existing Systems

- Focus only on computer vision processing
- No user-friendly mobile dashboard
- Limited real-time interaction

Target Audience

- Students
- Office workers
- Gamers
- Frequent screen users

Problem Insight: *Bridging the gap between computer vision analysis and user-centric real-time feedback.*

Objective of the Project

Main Objective

Develop a Flutter mobile application integrated with computer vision to monitor fatigue levels in real time.

Supporting Objectives

Live Status Display

Real-time rendering of current fatigue levels.

Eye Metric Tracking

Track blink counts and eye aspect ratios (EAR).

Fatigue Scoring

Calculate and display clinical fatigue scores.

Intelligent Alerts

Haptic and visual triggers when fatigue is detected.

Historical Data

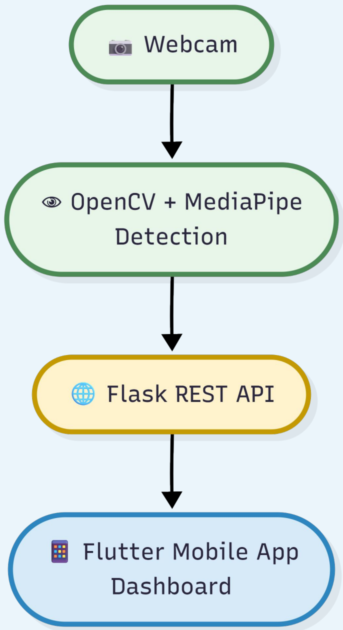
Access trend analysis and session reviews.

Expected Outcome: *A user-friendly application that improves awareness and helps users reduce eye fatigue.*

Proposed Solution

System Overview: The proposed system combines computer vision and a mobile application.

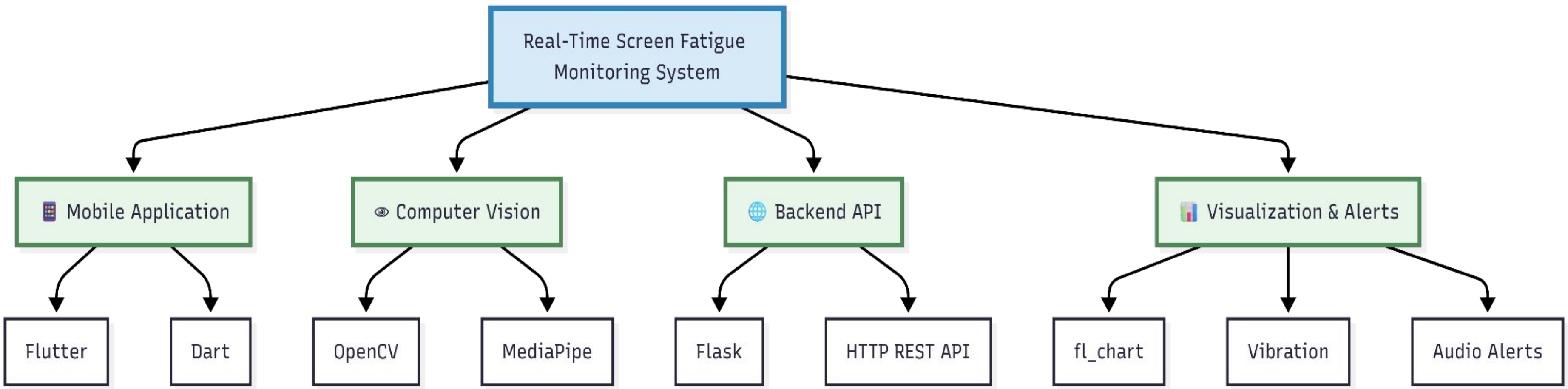
System Workflow



Key Benefits

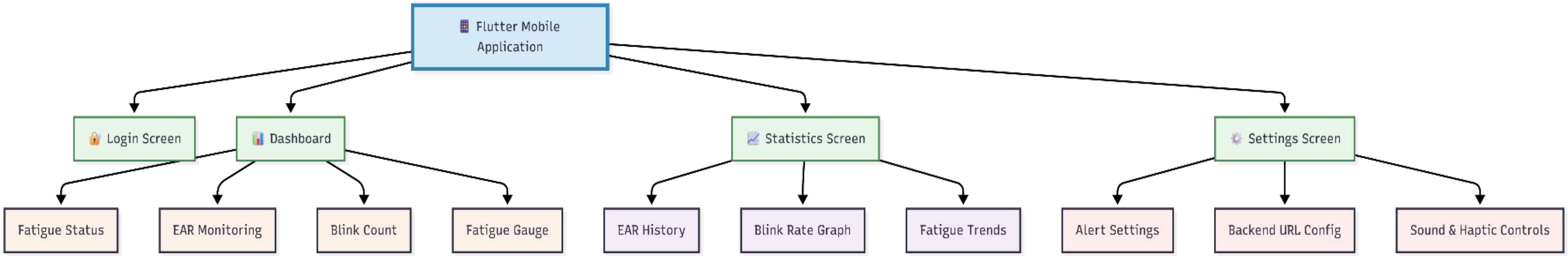
- **Continuous monitoring:** Seamless real-time tracking of fatigue levels.
- **Live updates:** Immediate data synchronization across the system.
- **Instant notifications:** Direct alerts to users when fatigue is detected.
- **Better user experience:** Intuitive interface for easy interaction.

Technology Stack



- **Flutter and Dart** are used to build a responsive and cross-platform mobile application.
- **OpenCV and MediaPipe** enable real-time face detection and eye landmark tracking.
- **Flask REST API** provides seamless communication between the detection system and the mobile app.
- **Charts and alert modules** help visualize fatigue data and notify users instantly.

System Architecture and Features



Dashboard Screen Features

Start / Stop
Toggle monitoring session with a single tap.

Fatigue Gauge
Animated radial gauge – green, amber, red severity zones.

Status Card
Plain-language summary: Alert, Normal, or Resting.

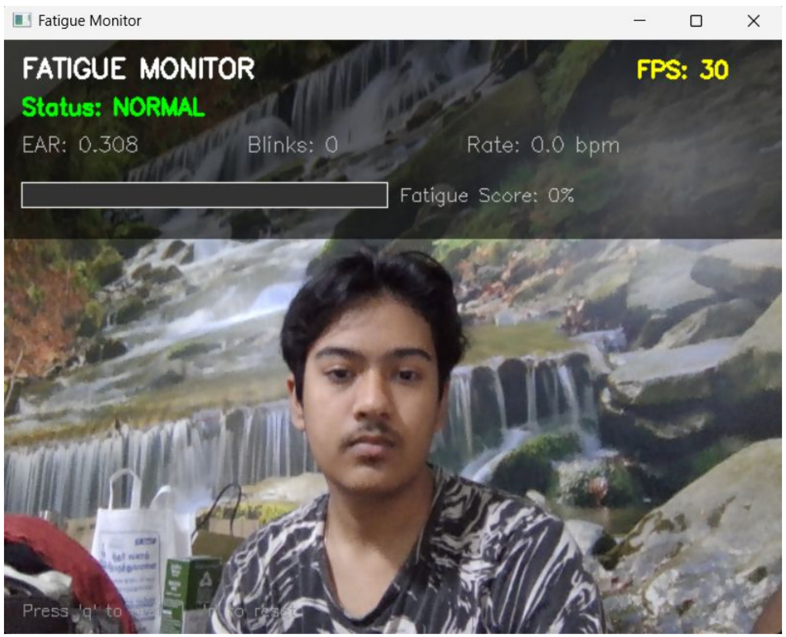
Blink Count
Cumulative and per-minute blink frequency display.

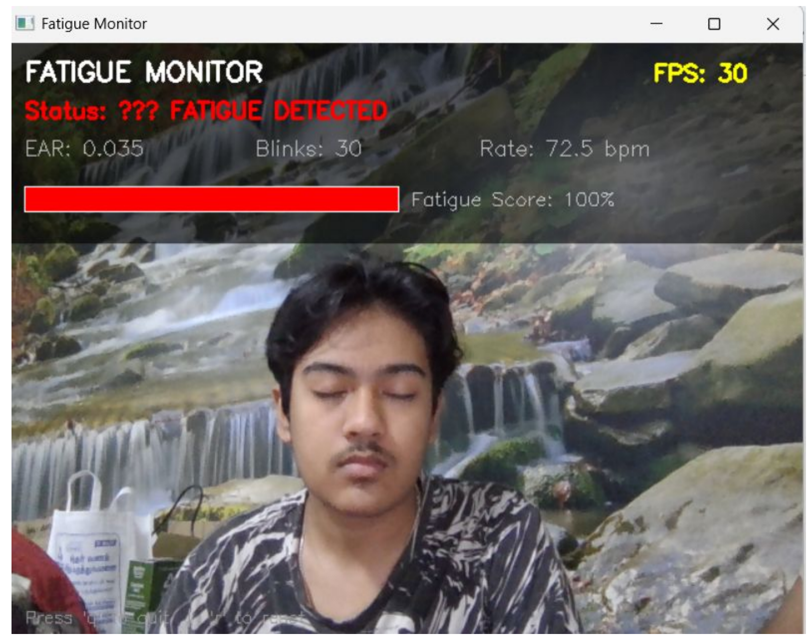
EAR Display
Live Eye Aspect Ratio value with threshold indicator.

Face Detection Status
Confirms active face lock – alerts if face leaves frame.

Face Detection using OpenCV

Capture Frame → Detect Face → Detect Landmarks → Extract Eye Region → Calculate EAR → Send Results to Flutter App





Output Screens / UI Screenshots

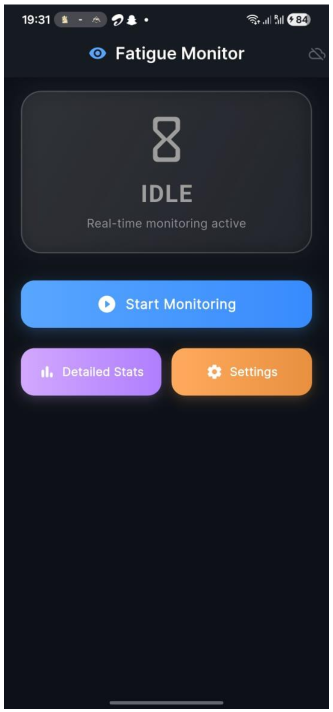


fig. Home Screen

fig. Fatigue Detected

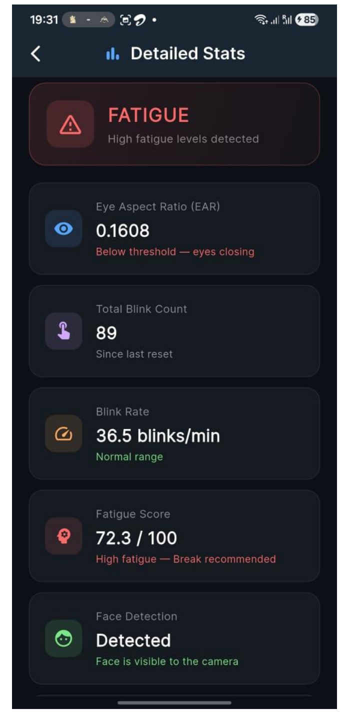


fig. Detailed Stats

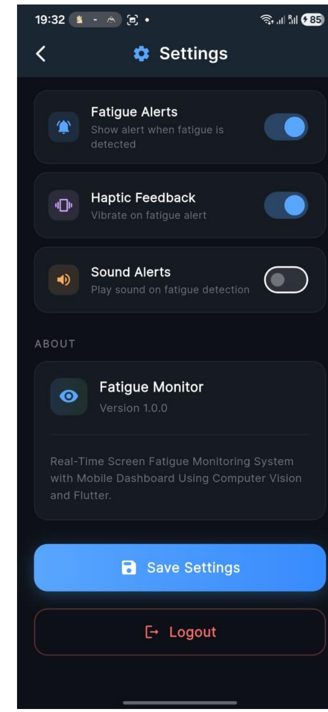


fig. Settings

Conclusion & Future Enhancement

Conclusion

The mobile application successfully combines computer vision and mobile technology to monitor user fatigue in real time.

Benefits:

- Detects fatigue early
- Provides live alerts
- Improves user awareness
- Easy and interactive interface

Future Enhancements

Planned upgrades to further improve system performance and user experience.

- AI prediction models
- Cloud synchronization
- Wearable device integration
- Personalized recommendations

Thank You…!